Domain and range and set notation

1 Write down all the elements of each of the following sets.

a $\{x|x \text{ is a natural number less than } 5\}$

b $\{x|x \text{ is an integer between } -8 \text{ and } -3 \text{ (not inclusive)}\}$

c The set positive factors of 42

2 Determine the symbol that would make each of the following statement true by choosing either $\in$ or $\notin$.

a $5 \text{ ▮ } \{2,5,6,9\}$

b $11 \text{ ▮ } \{4,8,12,14\}$

c $0 \text{ ▮ } \{0,3,4,5,6,8\}$

d $\{5\} \text{ ▮ } \{3,4,5,6,7\}$

e $0 \text{ ▮ } \emptyset$

3 Determine whether the following statements are true or false.

a $3 \in \{2,4,5,8\}$

b $1 \in \{9,8,6,5,1\}$

c $7 \notin \{6,5,4,2\}$

d $\{9,6,7,3,4\} = \{7,4,3,6,9\}$

e $\{7,4,5,8\} = \{5,4,0,8,7\}$

f $3 \in \{x|x \text{ is a rational number}\}$

g Some rational numbers are not positive.

4 Consider the set $A = \{2,4,6,8\}$.
Determine a possible set builder notation for A.

5 For each of the following relations:

i Determine the domain.

ii Determine the range.

iii Is this relation a function?

a $\{(8,5),(1,3),(3,4),(9,1),(2,9)\}$

b $\{(2,3),(3,8),(6,1),(8,7),(3,2)\}$

c $\{(4,4),(8,9),(2,8),(7,9),(3,1)\}$

d

x	1	6	3	8	2
y	3	2	7	1	2

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e

x	1	6	3	8	2
y	3	2	7	1	2

f

x	2	6	9	1	5
y	6	5	1	8	3

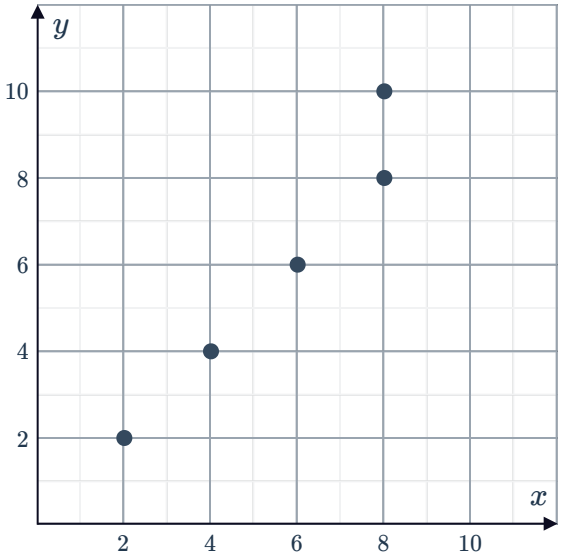
6 Consider the following graphs of relations:

i Determine the domain.

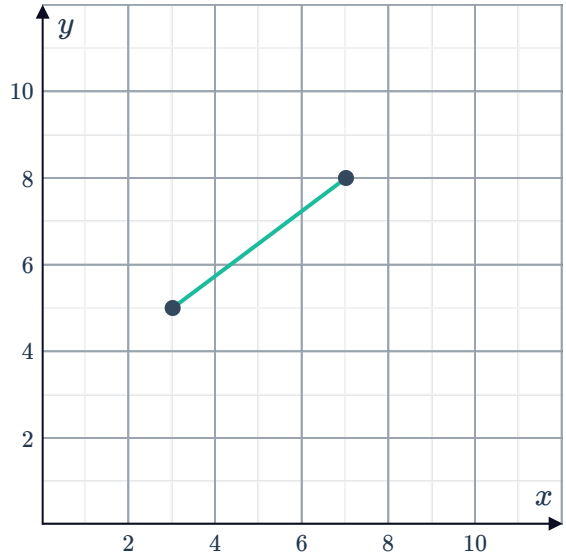
ii Determine the range.

iii Is this relation a function?

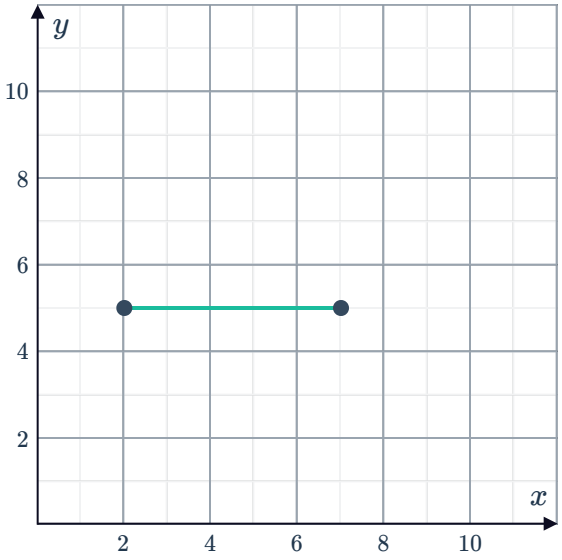
a



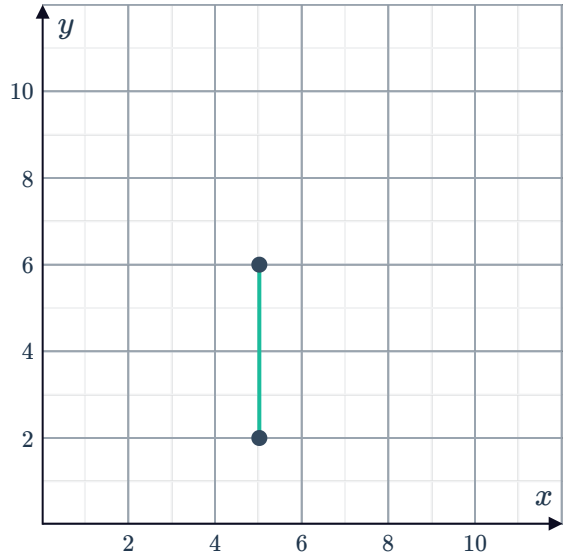
b



c

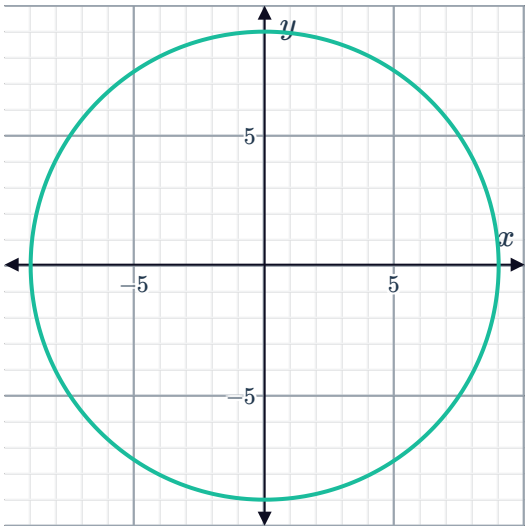


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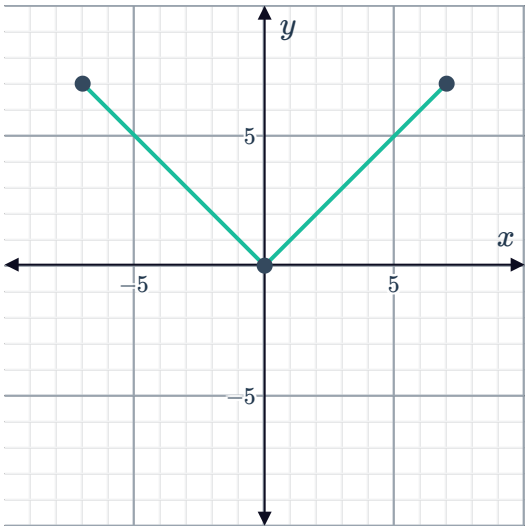


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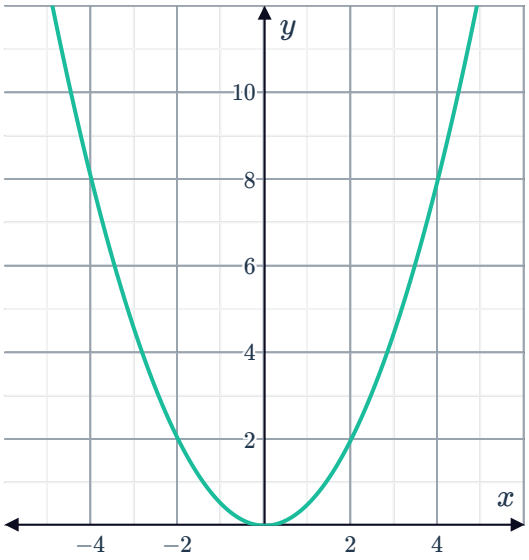
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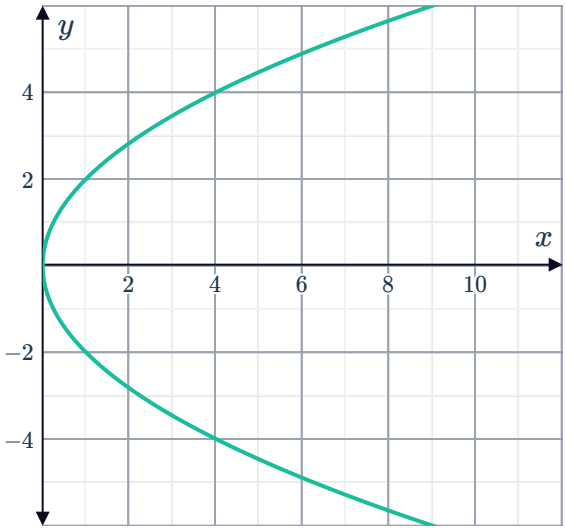
f



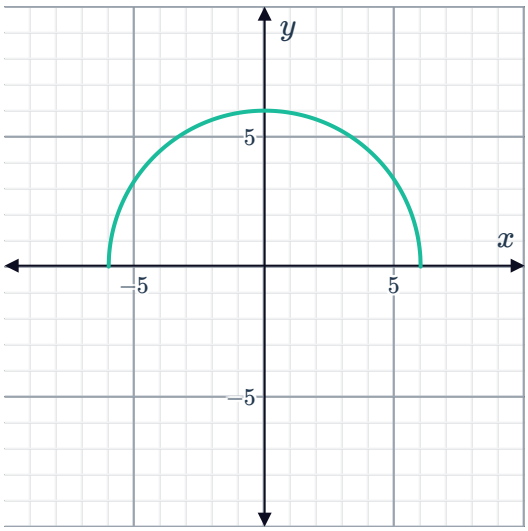
g



h



7 Consider the graph of the function $f(x) = \sqrt{36 - x^2}$.
State the domain of the function.



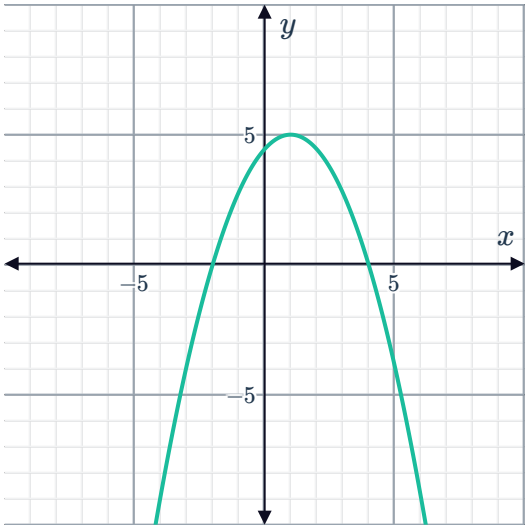
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8 Consider the following graphs of functions:

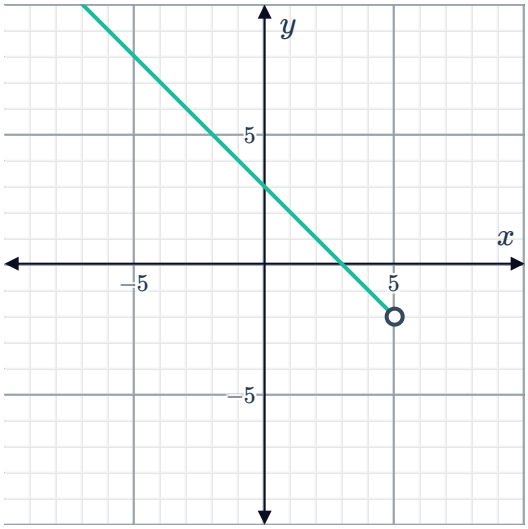
i Determine the domain of the function.

ii Determine the range of the function.

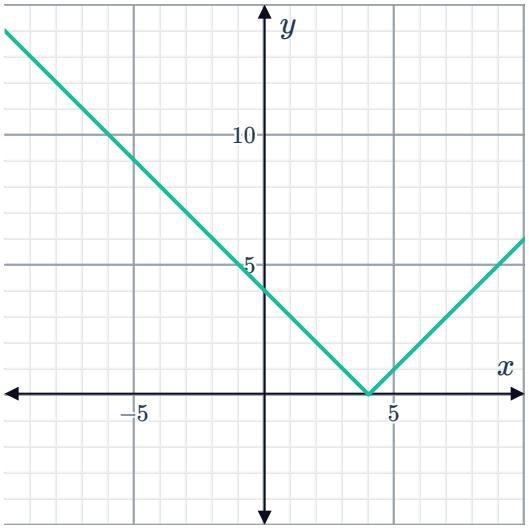
a



b



c

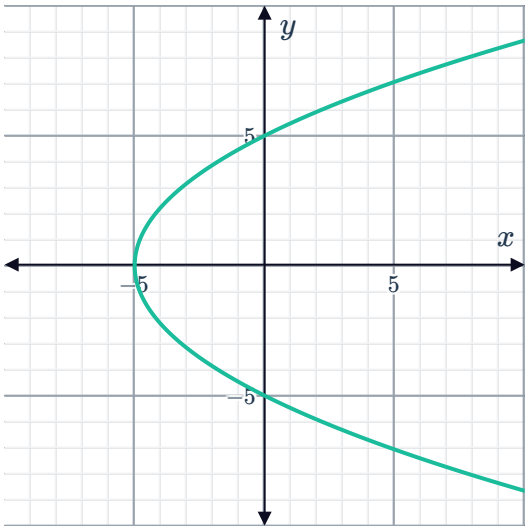


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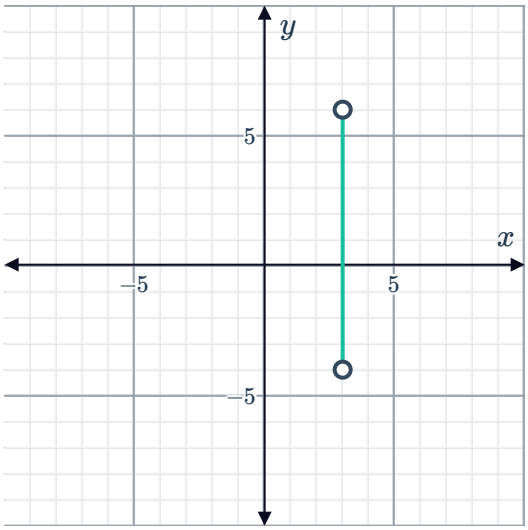
9 Determine whether the following graphs matches the given domain and range.

Domain	All real numbers
Range	$\{y \mid y \leq 0\}$

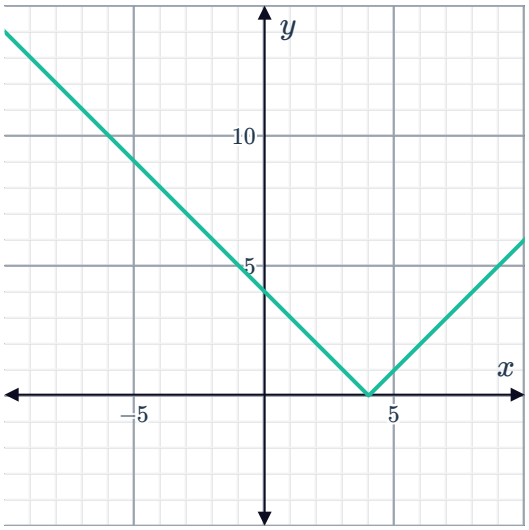
a



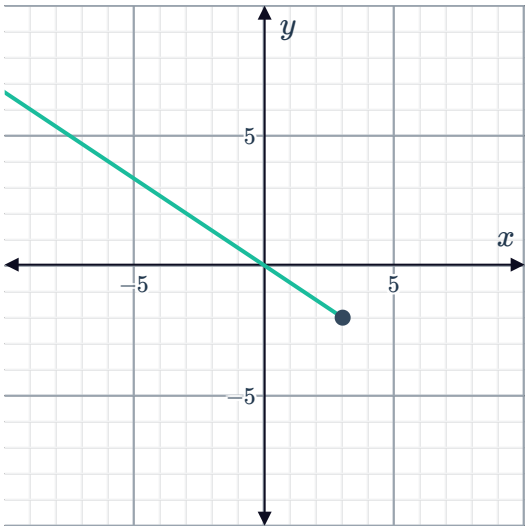
b



c



d

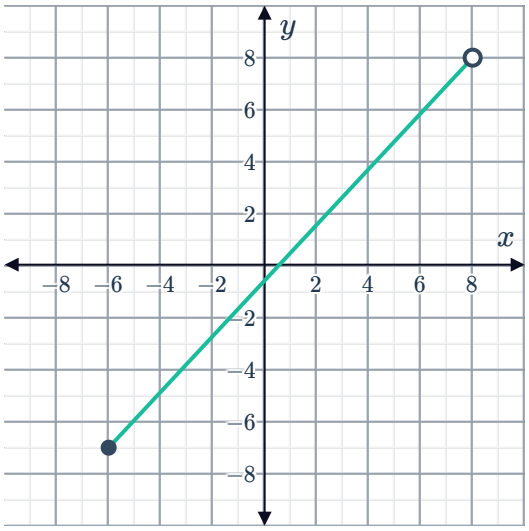


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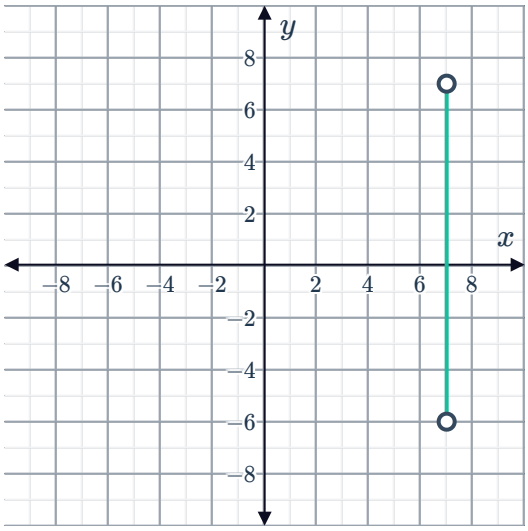
10 Determine whether the following graphs matches the given domain and range.

Domain	$\{x \mid x \leq 5\}$
Range	$\{y \mid y \geq -4\}$

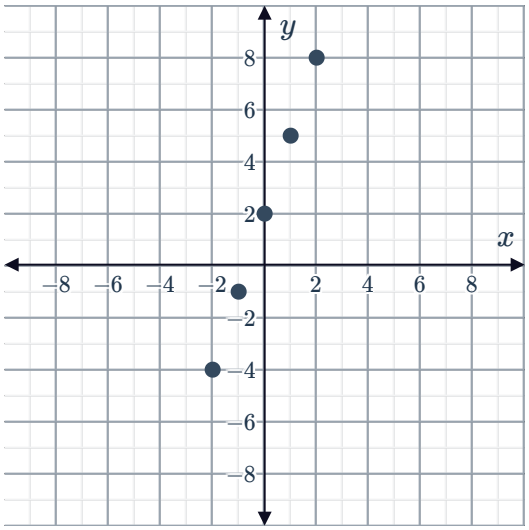
a



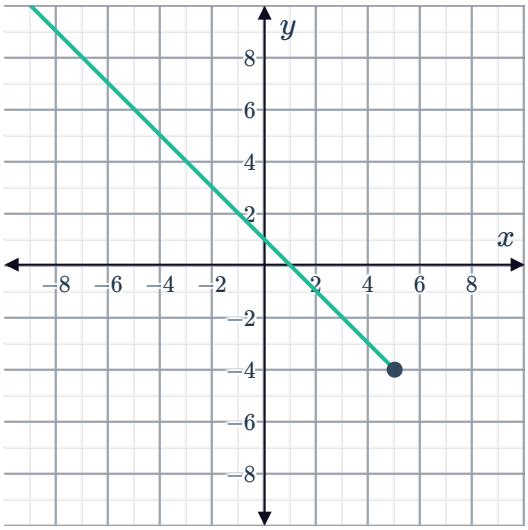
b



c



d

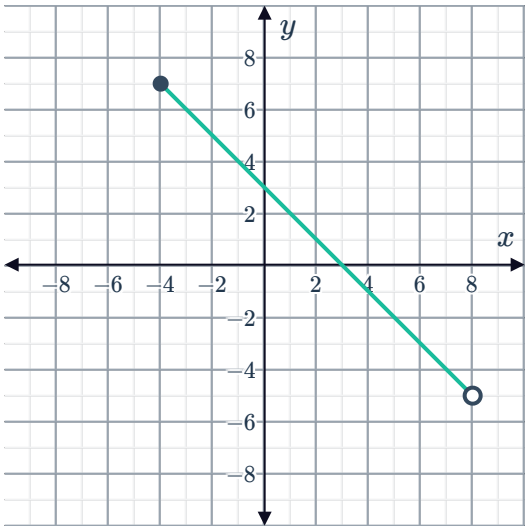


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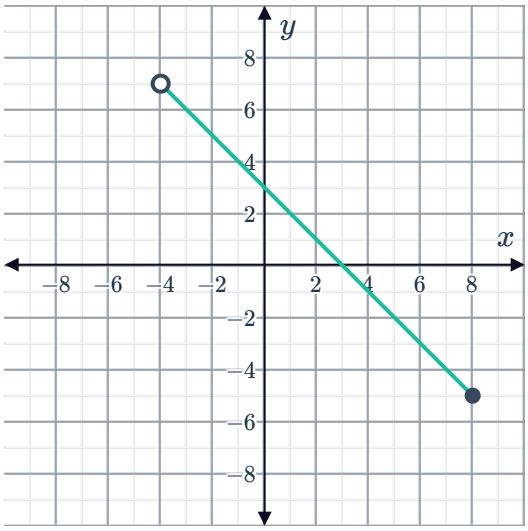
11 Determine whether the following graphs matches the given domain and range.

Domain	$\{x \mid -4 < x \leq 8\}$
Range	$\{y \mid -5 \leq y < 7\}$

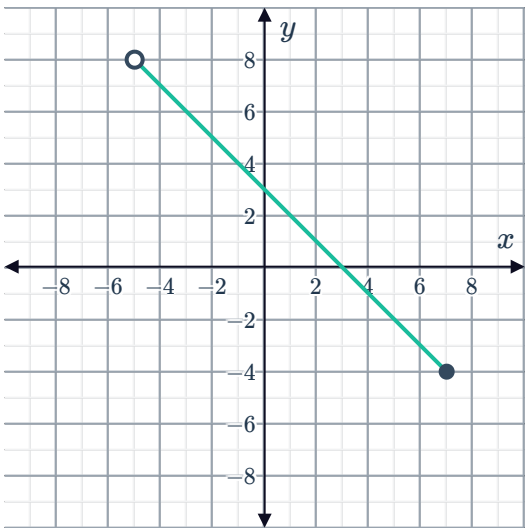
a



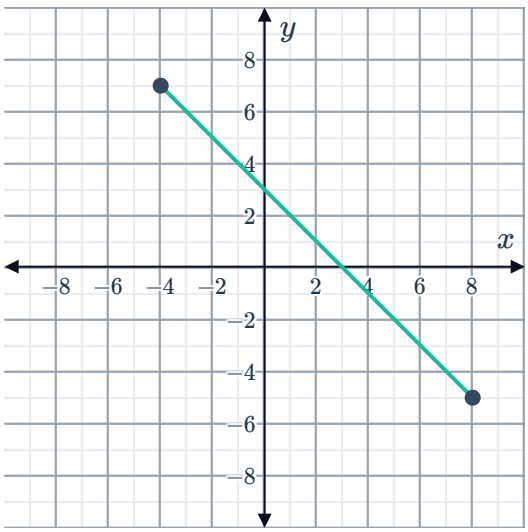
b



c



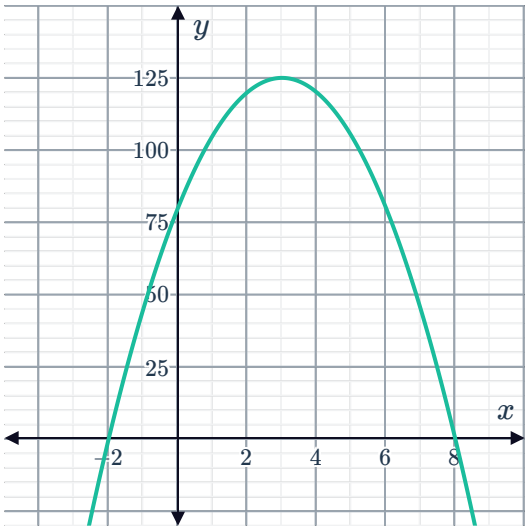
d



12 Consider the function $f(x) = 2x^2 - 10x + 12$.
Find the interval that represents the range of $f(x)$.

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13 The graph of a function is shown.



a Determine the range of the function.

b A ball is thrown from an apartment window in a high-rise building. The height of the ball above ground over time can be modelled by the function shown in the graph, where the ball is thrown at $x = 0$.
Determine the range of the function in this context.

14 The function $f(n)$ describes the maximum height of a ball dropped from a certain height after the nth bounce.
Determine whether the following could be the domain of the function.

a All non-negative rational numbers

b All non-negative integers

c All real numbers

d All integers

15 A whale jumps above the surface of the ocean and then dives back under. Its height above the surface can be sketched on a set of axes where x represents time and y represents distance above or below the surface.
Determine whether the following sets best represent the domain for this graph.

a The set of positive real numbers

b The set of positive integers

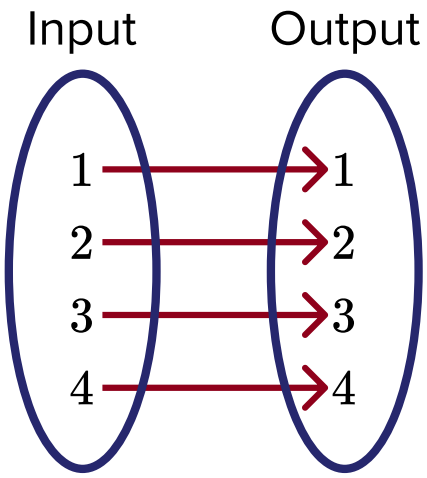
c The set of integers

d Real numbers

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Additional questions

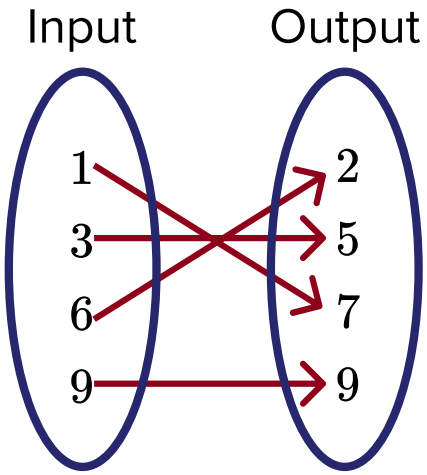
16 The following mapping diagram represents a relation between the input and the output.



a What is the domain of the relation?

b What is the range of the relation?

17 The following mapping diagram represents a relation between the input and the output.

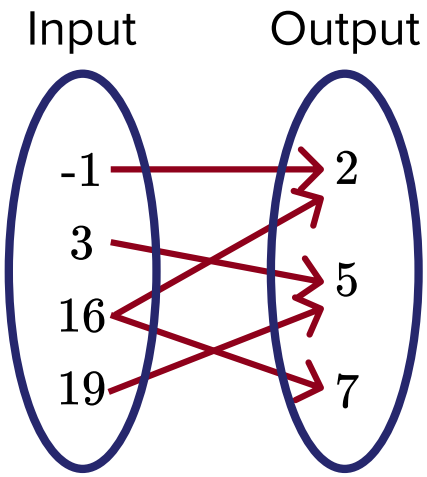


a What is the domain of the relation?

b What is the range of the relation?

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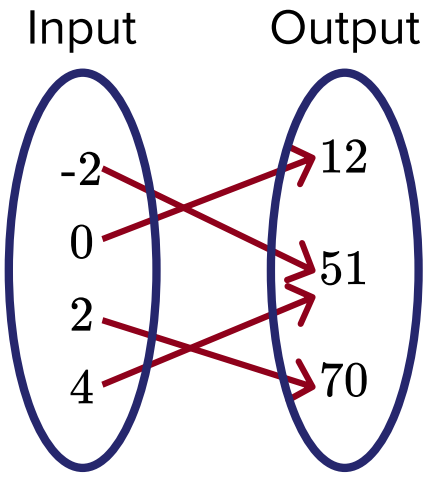
18 The following mapping diagram represents a relation between the input and the output.



a What is the domain of the relation?

b What is the range of the relation?

19 The following mapping diagram represents a relation between the input and the output.



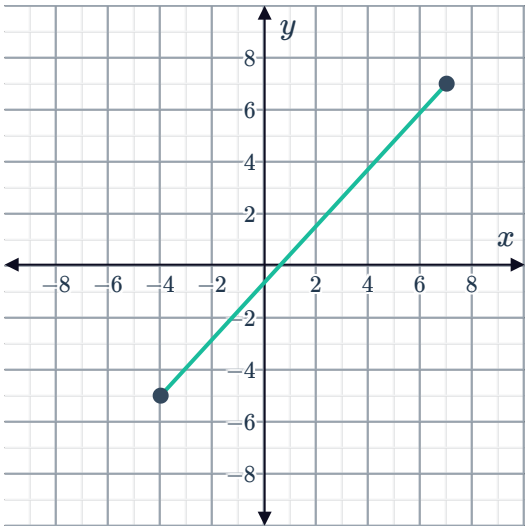
a What is the domain of the relation?

b What is the range of the relation?

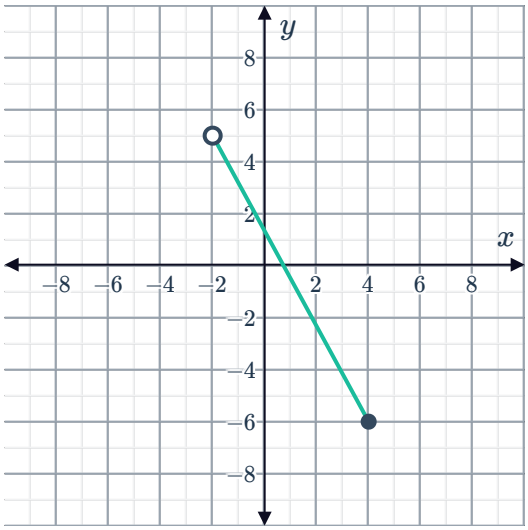
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20 State the domain and range for each of the following graphs:

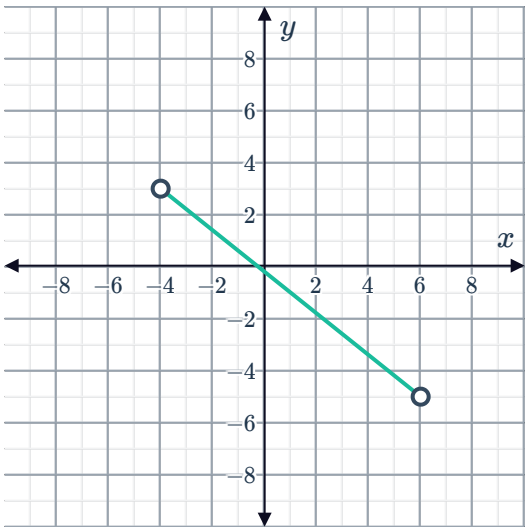
a



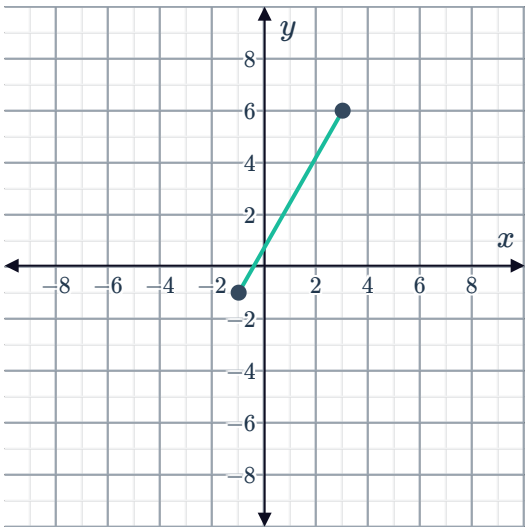
b



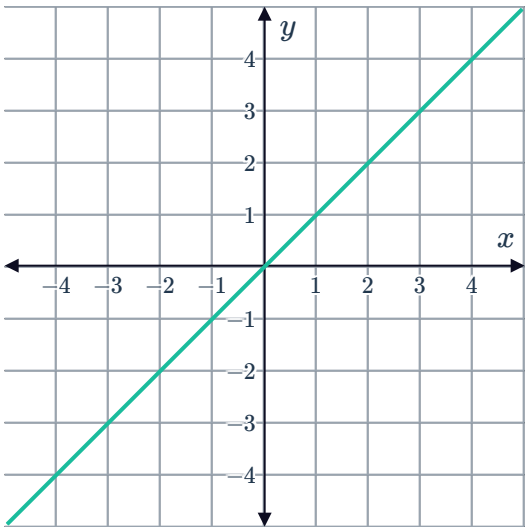
c



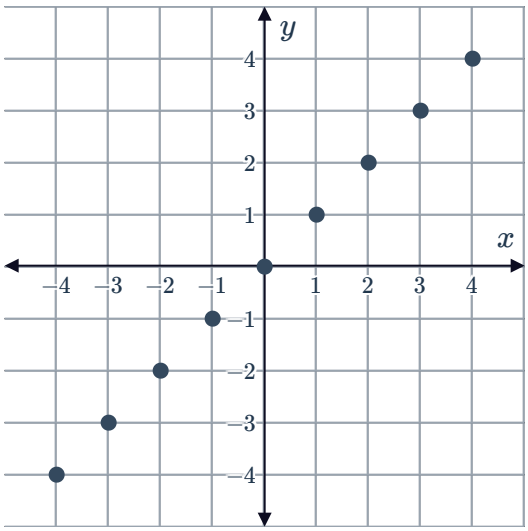
d



e

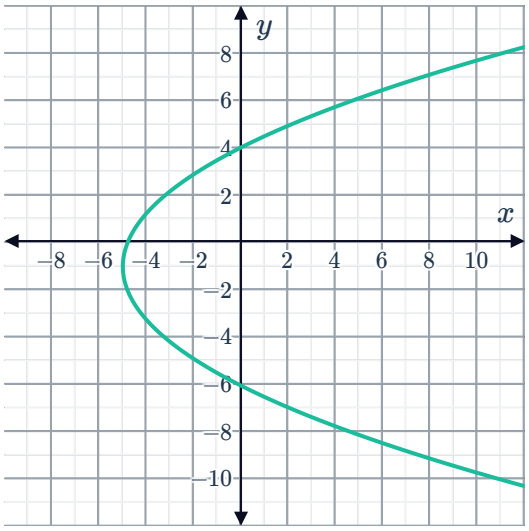


f

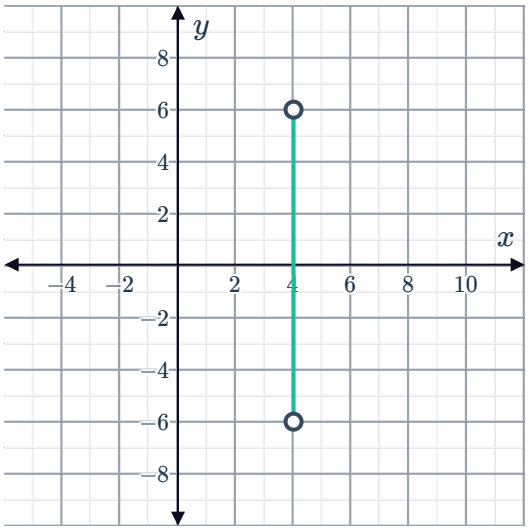


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g

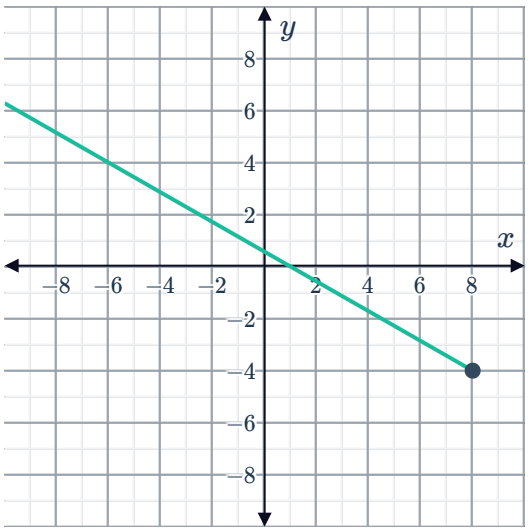


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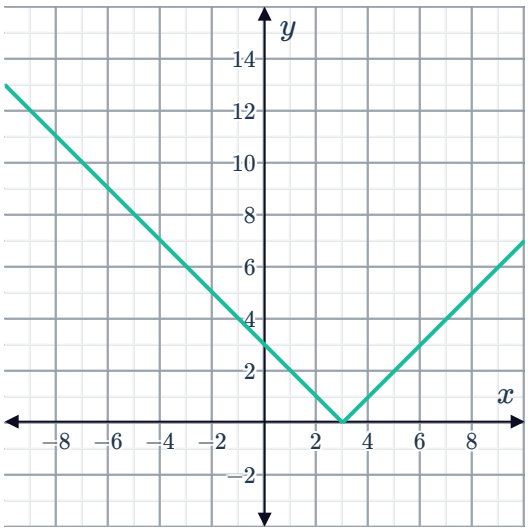


21 Determine for each graph whether or not it has a domain consisting of all the real numbers. If it does not, state the domain.

a



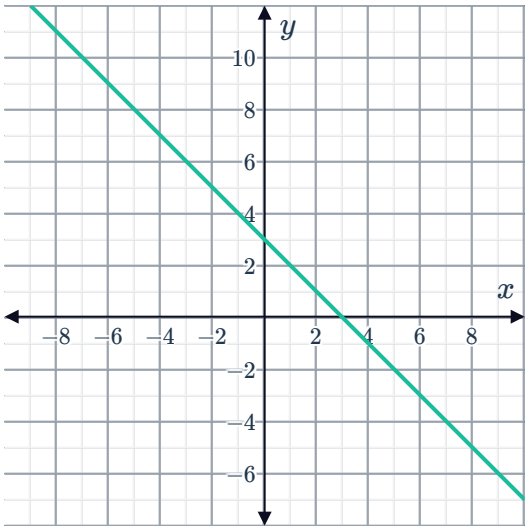
b



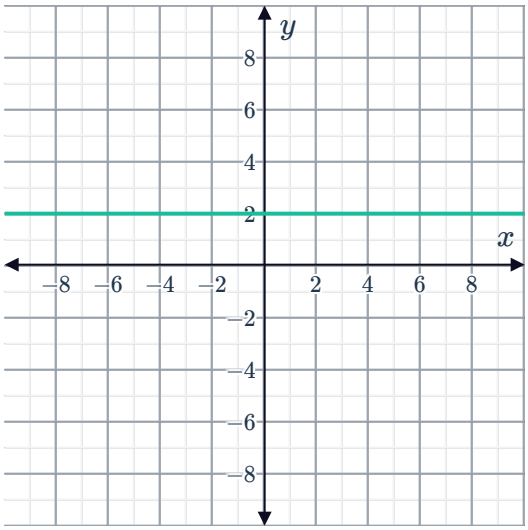
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22 Determine for each graph whether or not it has a range consisting of all the real numbers. If it does not, state the range.

a



b



23 State the domain and range for the following scenario.

a Uma earns between $\$200$ and $\$450$ selling between 35 and 50 burritos, inclusive.

b Jermaine starts reading his book from page 112. Over the course of 17 days, he reads up to and including page 237.

c Brigid Kosgei completes a 26.2 mile marathon in 134 minutes.

24 A comet is travelling through the solar system. It passes by the Earth before curving around the Sun and heading back out into deep space. The distance of the comet from Earth, y, is tracked by satellite telescopes and expressed as a function of time, x, since the comet was first spotted.

a State the domain of this function.

b The satellites detect that, at its closest point, the comet was a distance of $0.14\ \text{AU}$ (astronomical units) from Earth. State the range of the function.

25 A function describing the strictly increasing relation between temperature and a person's average resting heart rate has a domain of $50 \leq T \leq 105$ (Fahrenheit) and a range of $40 \leq f(T) \leq 90$ (beats per minute), based on experimental results.

a Determine the average resting heart rate at a temperature of 50°F.

b State the temperature that is expected if a person has an average resting heart rate of 90 beats per minute.

c Explain whether or not the function output $f(10)$ is a reliable estimate for the scenario.

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